

# LeetCode 1021 — Remove Outermost Parentheses

**Language:** Java

## Question

Given a valid parentheses string **s**, divide it into its primitive valid-parentheses strings. Remove the outermost pair of parentheses from every primitive string and return the resulting string.

## Example

**Input:** **s** = "()()()")"

**Primitive decomposition:** "()()")" + "()"")"

**After removing outer parentheses:** ")()")" + ")()"")" = ")()()")"

## Java Solution

```
class Solution {
    public String removeOuterParentheses(String s) {
        StringBuilder result = new StringBuilder();
        int balance = 0;

        for (char c : s.toCharArray()) {
            if (c == '(') {
                if (balance > 0) {
                    result.append(c);
                }
                balance++;
            } else {
                balance--;
                if (balance > 0) {
                    result.append(c);
                }
            }
        }

        return result.toString();
    }
}
```

## Simple Logic

Use **balance** to count the currently open parentheses. For an opening parenthesis, add it only when balance is already greater than 0. For a closing parenthesis, decrease balance first and add it only when balance is still greater than 0. This skips the first '(' and last ')' of every primitive string.

## Complexity

**Time:** O(n)

**Space:** O(n), for the output string.